

GENERAL PROJECTILES

17/11/25

Take g as 9.8 m s^{-2} unless instructed otherwise.

1. A particle is projected from an origin O with a velocity with horizontal component 4 m s^{-1} to the right and vertical component 13 m s^{-1} upward. Find the position of the particle 2 seconds after projection and the distance the particle is then from O . (Take $g = 10 \text{ m s}^{-2}$)
2. A particle is projected from an origin O with a velocity with horizontal component 4 m s^{-1} to the right and vertical component 11 m s^{-1} upward. It passes through a point P with position $(8, x)$. Taking $g = 10 \text{ m s}^{-2}$, find the time taken for the particle to reach P from O and the value of x .
3. A particle is projected from a point on level ground such that its initial velocity is 56 m s^{-1} at an angle of 30° above the horizontal. Find:
 - a. The time taken for the particle to reach its maximum height
 - b. The maximum height
 - c. The time of flight
 - d. The horizontal range of the particle
4. Find the horizontal range of a particle projected with a velocity of 35 m s^{-1} at an angle of $\arcsin(4/5)$ above the horizontal.
5. A ball is projected from horizontal ground and has an initial velocity of 20 m s^{-1} at an angle of $\arctan(7/24)$ above the horizontal. When the ball is travelling horizontally it hits a vertical wall. How high above ground level does the impact occur?
6. Take g as 10 m s^{-2} . A particle is projected from a horizontal plane and has an initial velocity of 50 m s^{-1} at an angle of $\arctan(4/3)$ above the horizontal. For how long is the particle at least 60m above the level of the plane?
7. A particle is projected from an origin O with initial velocity 44 m s^{-1} at an angle of 30° above the horizontal. Find expressions for the horizontal and vertical components of velocity and displacement of the particle from O , t seconds later.
8. A particle is projected at a velocity of 25 m s^{-1} at an angle of 30° above the horizontal. Find the horizontal and vertical components of the particle's velocity $2\frac{1}{2}$ seconds after projection. Hence find the speed and direction of motion of the particle at that time.
9. Take g as 10 m s^{-2} . A stone is thrown from the edge of a vertical cliff and has an initial velocity of 26 m s^{-1} at an angle of $\arctan(5/12)$ below the horizontal. The stone hits the sea at a point level with the base of the cliff and 72m from it. Find the height of the cliff and the time for which the stone is in the air.
10. A particle is projected from a horizontal plane and has an initial velocity of 49 m s^{-1} at an angle of 30° above the horizontal. For how long is the particle at least 19.6m above the level of the plane?
11. Take g as 10 m s^{-2} . A stone is thrown from the top of a vertical cliff, 100m above sea level. The initial velocity of the stone is 13 m s^{-1} at an angle of $\arctan(5/12)$ above the horizontal. Find the time taken for the stone to reach the sea and its horizontal distance from the cliff at that time.